

PLSC 476: Empirical Legal Studies

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Analyzing Text: Goals

Humans:

- Good at: Meaning, subtlety (irony, sarcasm, subtle negation, etc.), context, tone, etc.
- Bad at: Doing things quickly and consistently.

Computers:

- Good at: Doing things quickly and consistently.
- Bad at: Meaning, subtlety (irony, sarcasm, subtle negation, etc.), context, tone, etc.

Key: Use computers / humans for what they're good at...

Example: SEC “Litigation Releases”

- Short (2-4 paragraph) summaries of civil litigation matters involving the Securities and Exchange Commission (SEC)
- Issues include securities (and other) fraud, insider trading, illegal stock sales, etc.
- Summaries of settled suits, litigation / trial outcomes, etc. plus charges filed
- Available from 1995-2025; we'll focus on 2024...
- Links are available [here](#)
- There are a lot of things you can do with these data; see, e.g., [here](#).



ENFORCEMENT & LITIGATION

Litigation Releases

Administrative Proceedings

Administrative Law Judge
Orders and Initial Decisions

Trading Suspensions

Distributions to Harmed
Investors

Receiverships

Electronic Filings in
Administrative Proceedings
(eFAP)

Rules of Practice

Applications for Reentry

Reports of Investigations

[Home](#) / [Enforcement and Litigation](#) / [Litigation Releases](#) / [Auerbach et al.](#)

Auerbach et al.

U.S. SECURITIES AND EXCHANGE COMMISSION Litigation Release No. 26170 / November 18, 2024

Securities and Exchange Commission v. Auerbach et al., No. 1:19-cv-5631 (E.D.N.Y. filed Oct. 4, 2019)

SEC Obtains Final Judgment Against Defendant for Receiving Kickbacks in Bribery Scheme

On November 15, 2024, the Securities and Exchange Commission obtained a final judgment against defendant Richard Brown for accepting kickbacks in an alleged scheme to defraud investors.

The SEC's complaint, filed in the Eastern District of New York on October 4, 2019, alleged that from approximately July 2014 through October 2015, Brown, then a registered stockbroker, accepted cash bribes from the CEO of Nxt-ID, Inc. and another defendant to purchase more than \$750,000 worth of Nxt-ID, Inc. common stock in his customers' accounts. Brown allegedly made the purchases without informing his customers that he was receiving kickbacks in connection with the purchases.

RESOURCES

- [Final Judgment - Richard Brown](#)

Source Code...

```
view-source:https://www.sec.gov/enforcement-litigation/litigation-releases/lr-26170

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Litigation Release No. 26170 / November 18, 2024</h2>

<p></p>

<h3><i>Securities and Exchange Commission v. Auerbach et al.</i></i> </i>No. 1:19-cv-5631 (E.D.N.Y. filed Oct. 4, 2019)</h3>

<p></p>

<h3>SEC Obtains Final Judgment Against Defendant for Receiving Kickbacks in Bribery Scheme</h3>

<p></p>

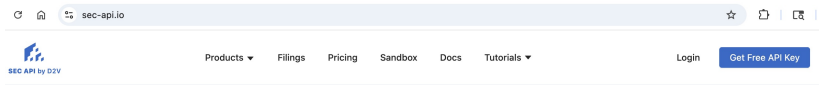
<p>On November 15, 2024, the Securities and Exchange Commission obtained a final judgment against defendant Richard Brown for accepting kickbacks
<p>The SEC's complaint, filed in the Eastern District of New York on October 4, 2019, alleged that from approximately July 2014 through October 20
<p>Without admitting or denying the SEC's allegations, Brown consented to the entry of a final judgment permanently enjoining him from violations
<p>The litigation was led by Mary Kay Dunning and Lindsay S. Moilanen and supervised by Tejal D. Shah and Preethi Krishnamurthy.</p>
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```

We could...

- ...scrape the data directly from the SEC website
 - Pros: Simple, complete, retains context
 - Cons: Highly secure vs. DDOS, seriously rate-limited
- ...work with the SEC's [RSS feed](#) (clunky...)
- ...use a third-party aggregator's API



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18+ million

SEC EDGAR filings

Access any SEC dataset with our simple-to-use APIs

Try it in our sandbox

A Function to Read the API Data

```
Fetch_SEC_Litigation_2024 <- function(api_key) {  
  
  # API endpoint  
  url <- "https://api.sec-api.io/sec-litigation-releases"  
  
  # Initialize results  
  all_data <- list()  
  from <- 0  
  size <- 50  
  total_fetched <- 0  
  .  
  .  
  .  
  
  # Convert to dataframe  
  lit_releases_df <- bind_rows(lapply(all_data, function(x) {  
    data.frame(  
      id = x$id %||% NA_character_,  
      release_number = x$releaseNo %||% NA_character_,  
      date = x$releasedAt %||% NA_character_,  
      url = x$url %||% NA_character_,  
      title = x$title %||% NA_character_,  
      subtitle = x$subTitle %||% NA_character_,  
      summary = x$summary %||% NA_character_,  
      tags = paste(unlist(x$tags), collapse = "; "),  
      entities = paste(sapply(x$entities, function(e) e$name %||% ""), collapse = "; "),  
      has_settlement = x$hasAgreedToSettlement %||% NA,  
      has_penalty = x$hasAgreedToPayPenalty %||% NA,  
      stringsAsFactors = FALSE  
    )  
  })))  
  
  # Convert date to proper format  
  lit_releases_df$date <- as.POSIXct(lit_releases_df$date,  
                                     format = "%Y-%m-%dT%H:%M:%S",  
                                     tz = "America/New_York")  
  
  cat(sprintf("Successfully fetched %d litigation releases from 2024\n",  
             nrow(lit_releases_df)))  
  
  return(lit_releases_df)
```


We...

- ...must first get an API key from the website (free, good for 100 queries)
- ...then run some simple code:

```
> API_KEY <- "4a242ecbd6e6e5f839485b5e56d7edef8c91643b05b8ace3b13d3"
> SEC2024.df <- Fetch_SEC_Litigation_2024(API_KEY)
Fetching SEC litigation releases from 2024 via API...
Fetched 50 of 288 records
Fetched 100 of 288 records
Fetched 150 of 288 records
Fetched 200 of 288 records
Fetched 250 of 288 records
Fetched 288 of 288 records
Successfully fetched 288 litigation releases from 2024
```

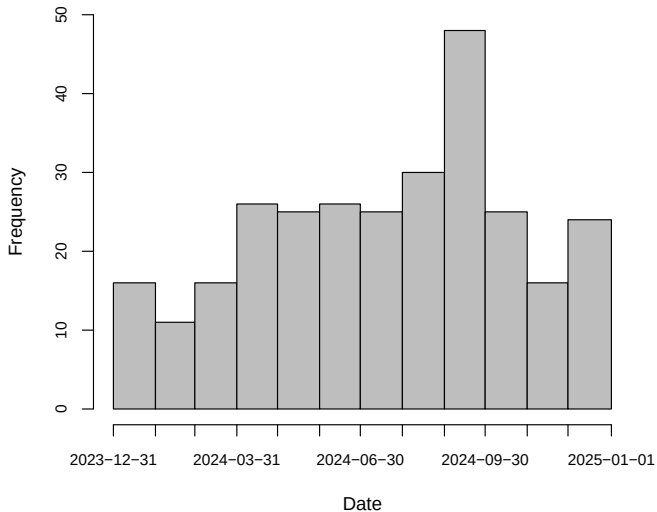
- ...which yields a data frame with 288 rows and eleven columns...

The Data

```
> summary(SEC2024.df)
```

id	release_number	date	
Length:288	Length:288	Min. :2024-01-03 15:51:50	
Class :character	Class :character	1st Qu.:2024-05-01 14:36:50	
Mode :character	Mode :character	Median :2024-07-30 19:22:44	
		Mean :2024-07-18 14:50:39	
		3rd Qu.:2024-09-27 18:07:08	
		Max. :2024-12-31 01:53:13	
url	title	subtitle	summary
Length:288	Length:288	Length:288	Length:288
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character
tags	entities	has_settlement	has_penalty
Length:288	Length:288	Mode :logical	Mode :logical
Class :character	Class :character	FALSE:146	FALSE:144
Mode :character	Mode :character	TRUE :140	TRUE :142
		NA's :2	NA's :2

Litigation Releases By Date



→ Corpus → DTM

```
> SEC<-VCorpus(DataframeSource(SEC2024.df))

> SEC

<<VCorpus>>
Metadata: corpus specific: 0, document level (indexed): 13
Content: documents: 288

> SEC.DTM<-DocumentTermMatrix(SEC,
+                             control=list(removePunctuation=TRUE,
+                                           tolower=TRUE,
+                                           stopwords=TRUE,
+                                           removeNumbers=TRUE,
+                                           stemming=TRUE))

> rownames(SEC.DTM)<-SEC2024.df$release_number # release IDs...

> SEC.DTM

<<DocumentTermMatrix (documents: 288, terms: 1479)>>
Non-/sparse entries: 5765/420187
Sparsity           : 99%
Maximal term length: 18
Weighting          : term frequency (tf)
```

“Search”-Like Things: Fraud

```
> frauds<-SEC.DTM[,grepl("fraud",SEC.DTM$dimnames$Terms)]
```

```
> frauds
```

```
<<DocumentTermMatrix (documents: 288, terms: 4)>>
```

```
Non-/sparse entries: 184/968
```

```
Sparsity           : 84%
```

```
Maximal term length: 9
```

```
Weighting           : term frequency (tf)
```

```
> inspect(frauds)
```

```
<<DocumentTermMatrix (documents: 288, terms: 4)>>
```

```
Non-/sparse entries: 184/968
```

```
Sparsity           : 84%
```

```
Maximal term length: 9
```

```
Weighting           : term frequency (tf)
```

```
Sample             :
```

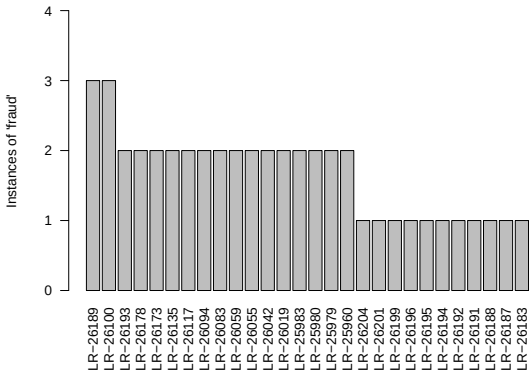
	Terms			
Docs	antifraud	defraud	fraud	fraudul
LR-26059	0	0	1	1
LR-26083	0	1	1	0
LR-26094	0	1	0	1
LR-26100	0	1	1	1
LR-26117	0	1	0	1
LR-26135	0	1	0	1
LR-26173	0	1	1	0
LR-26178	0	1	0	1
LR-26189	0	1	1	1
LR-26193	1	0	1	0

More Fraud...

```
> table(row_sums(frauds))
```

```
  0   1   2   3
122 149  15   2
```

```
# Highlight the documents with the greatest incidence of "fraud"...
```



Weighting (TF v. TF-IDF)

A standard *document-term matrix* has:

- $i \in 1 \dots N$ rows, corresponding to the N documents D in the corpus
- $j \in 1 \dots J$ columns, corresponding to the J unique terms T in the corpus
- Cell entries N_{ij} that represent the number of times term j appears in document i

Term frequency:

N_{ij} = The number of times term j appears in document i

Term frequency (normalized for document length):

$$TF_{ij} = \frac{N_{ij}}{\sum_{i=1}^J N_{ij}},$$

the fraction of all terms in document D_i that are term T_j .

Inverse document frequency (normalized):

$$IDF_j = \log_2 \frac{J}{J_j}$$

where J_j is the number of documents in which T_j occurs.

TF-IDF_{ij} is then simply $TF_{ij} \times IDF_j$

TF-IDF: Examples

Three “documents”:

$A = \{\text{red, blue, red}\}$
 $B = \{\text{green, blue, orange}\}$
 $C = \{\text{yellow, blue, yellow}\}$

Example one:

- In document A “red” appears twice ($TF_{ij} = 2$), and
- “red” is two of the three total terms in that document (normed $TF_{ij} = 2/3 = 0.67$)
- “red” appears in only one of the three documents ($IDF_i = \log_2[3/1] = 1.6$)
- The TF-IDF for “red” in document A is $0.67 \times 1.6 = 1.1$

Example two:

- In document C “blue” appears once ($TF_{ij} = 1$), and
- “blue” is one of the three total terms in that document (normed $TF_{ij} = 1/3 = 0.33$)
- “blue” appears in all three documents ($IDF_i = \log_2[3/3] = 0$)
- The TF-IDF for “blue” in document C is $0.33 \times 0 = 0$

In general:

- (Normalized) TF indicates the prevalence of a term in a document
- IDF reflects how common or rare the word is across documents
- IDF is thus a measure of the level of “informativeness” (or “document-specificity”) of a word
- TF-IDF is thus a measure of a term’s **“importance”** (in some respects)
 - TF-IDF is zero if a term does not appear in a document at all
 - The TF-IDF is also always zero for any term that appears in *all* documents in a corpus
 - Values of TF-IDF are generally < 1 , but need not be
 - Higher values of TF-IDF indicate more important / distinctive terms for that document

Making a TF-IDF Document-Term Matrix

```
> SEC.TFIDF<-weightTfIdf(SEC.DTM) #TF-IDF weighting
```

```
> SEC.TFIDF
```

```
<<DocumentTermMatrix (documents: 288, terms: 1479)>>
```

```
Non-/sparse entries: 5765/420187
```

```
Sparsity : 99%
```

```
Maximal term length: 18
```

```
Weighting : term frequency - inverse document frequency (normalized) (tf-idf)
```

```
> inspect(SEC.TFIDF)
```

```
<<DocumentTermMatrix (documents: 288, terms: 1479)>>
```

```
Non-/sparse entries: 5765/420187
```

```
Sparsity : 99%
```

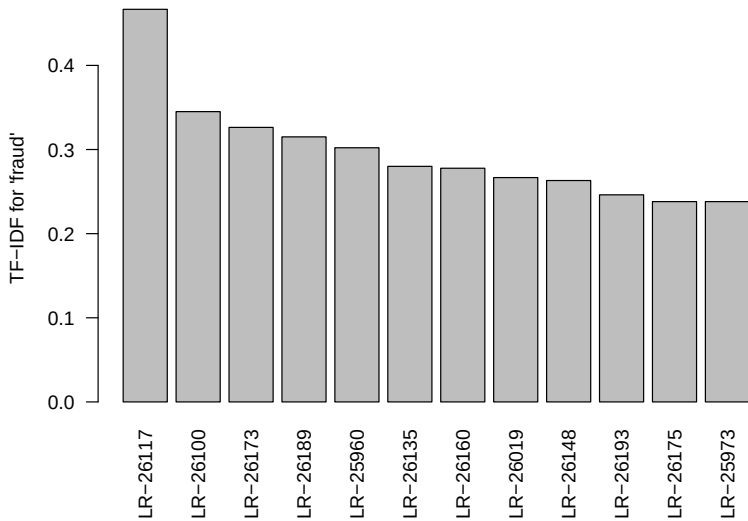
```
Maximal term length: 18
```

```
Weighting : term frequency - inverse document frequency (normalized) (tf-idf)
```

```
Sample :
```

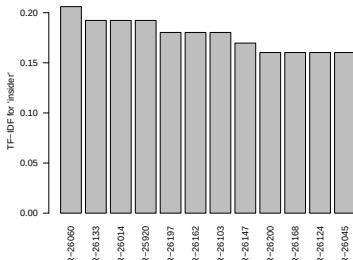
Docs	Terms									
	final	fraud	investor	judgment	million	obtain	penalti	result	secur	trade
LR-25942	0	0.00000000	0.00000000	0	0.00000000	0.06389066	0	0.00000000	0.00000000	0.00000000
LR-26021	0	0.00000000	0.13865045	0	0.06050684	0.00000000	0	0.00000000	0.00000000	0.00000000
LR-26029	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0.00000000
LR-26108	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0.10396736
LR-26109	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0.00000000
LR-26177	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0	0.00000000	0.07048661	0.00000000
LR-26194	0	0.06859846	0.08087943	0	0.07059132	0.00000000	0	0.00000000	0.00000000	0.00000000
LR-26195	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0	0.00000000	0.07929744	0.00000000
LR-26202	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0.00000000
LR-26206	0	0.00000000	0.00000000	0	0.00000000	0.00000000	0	0.0566058	0.00000000	0.08168864

Dozen Top "Fraud" Documents (by TF-IDF)



Another Term: “insider”

```
> IT<-SEC.TFIDF[,grepl("insider",SEC.DTM$dimnames$Terms)]  
  
> inspect(IT)  
<<DocumentTermMatrix (documents: 288, terms: 1)>>  
Non-/sparse entries: 39/249  
Sparsity          : 86%  
Maximal term length: 5  
Weighting          : term frequency - inverse document frequency (normalized) (tf-idf)  
Sample            :  
      Terms  
Docs      insider  
LR-25920  0.1923015  
LR-26014  0.1923015  
LR-26060  0.2060373  
LR-26103  0.1802827  
LR-26133  0.1923015  
LR-26147  0.1696778  
LR-26162  0.1802827  
LR-26168  0.1602513  
LR-26197  0.1802827  
LR-26200  0.1602513
```



Moving Beyond...

- More advanced search
- Correlations among terms within documents
- Measurement models (e.g., for combining similar or related terms)
- Visualizing “outlier” observations
- Various natural language processing tools:
 - Entity recognition
 - Sentiment analysis
 - Topic models
 - LLMs (for all of the above, and more)